

SMART ENVIRONMENTAL MONITOR

DecodeLabs IoT Internship – Project 1

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1. Project Overview

A Smart Environmental Monitor developed using an ESP32, DHT22 temperature and humidity sensor, and a 16×2 I2C LCD. The system reads environmental values and displays them on the LCD and Serial Monitor.

2. Objectives

- Measure temperature and humidity.
- Display readings on a 16×2 I2C LCD.
- Monitor readings through the Serial Monitor.
- Demonstrate an ESP32-based IoT sensing system.

3. Components Used

Component	Purpose
ESP32 Development Board	Main controller
DHT22 Sensor	Temperature and humidity sensing
16×2 I2C LCD	Display readings
Jumper Wires	Connections
USB Cable / Power	Programming and power

4. Pin Connections

DHT22: DATA → GPIO 4; VCC → 3.3V; GND → GND.

I2C LCD: SDA → GPIO 21; SCL → GPIO 22; VCC → supply; GND → GND.

Smart Environmental Monitor - Circuit Diagram

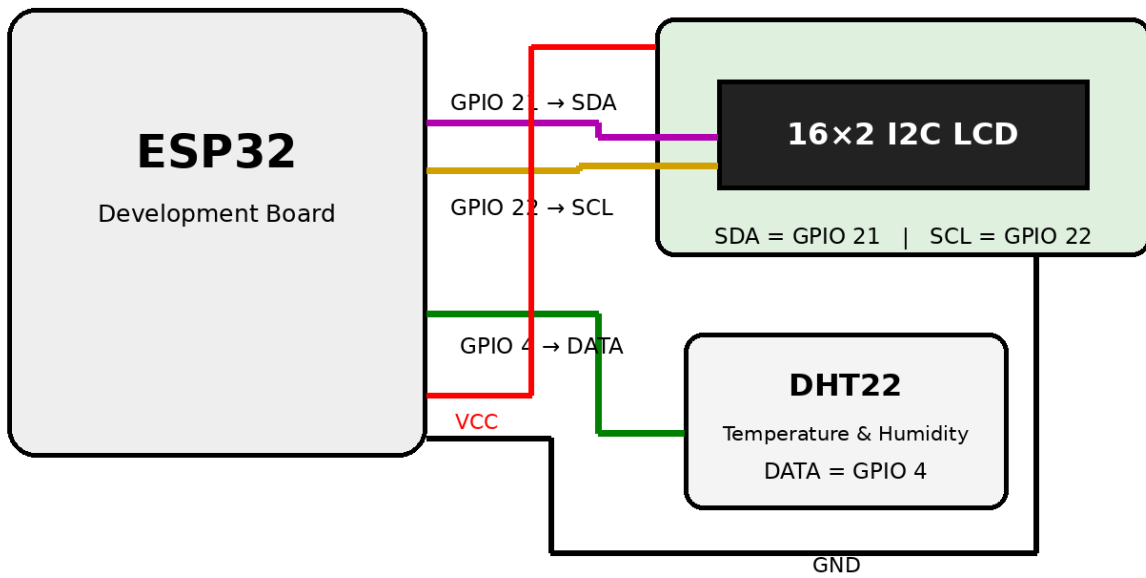


Figure 1: Circuit diagram.

5. Working Principle

The DHT22 senses temperature and humidity. The ESP32 reads the sensor data, then sends the values to the I2C LCD and Serial Monitor.

6. Software

Arduino IDE with ESP32 board support, DHT library and LiquidCrystal I2C library. The provided code uses DHT data on GPIO 4 and an I2C LCD address of 0x27.

7. Expected Output

Temperature and humidity values should appear on the LCD and Serial Monitor.

Arduino Serial Monitor - EXAMPLE OUTPUT

Replace this with your actual Serial Monitor screenshot.

```
Smart Environmental Monitor
```

```
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```

```
Temperature: 25.4 °C
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```
Humidity: 58.2 %
```

```
Temperature: 25.6 °C
```

```
Humidity: 58.0 %
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```
Temperature: 25.7 °C
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```
Humidity: 57.8 %
```

Figure 2: Example Serial Monitor output — replace with your actual screenshot.

8. Conclusion

The project demonstrates basic environmental sensing, ESP32 processing, I2C LCD display and serial monitoring in an IoT application.

Important: The serial_output.png included here is an example, not a real measurement. Replace it with your actual Serial Monitor screenshot before submission.